

Illegal gold mining in the Amazon: assignment brief

Data for Good, Lessons 3 to 5 · Use any software you like · Starter code in Python and R

Practice partner. The Frankfurt Zoological Society (FZS) is a conservation organization that works to protect wilderness areas in the Amazon, including in Peru and Colombia. Illegal gold mining is one of the main threats to these areas and to the people who live in them.

Your task. Each group answers **one** of five guiding questions, using publicly available data. The question cards show which question your group has. In Lesson 5 you present your work to Christof Schenck, Executive Director of FZS: a 10-minute presentation, followed by 5 minutes of questions, and a two-page brief. Start your presentation with two fun facts about your group's animal.

Deliverables

When	What
Today, by end of day	Team plan , half a page (template in the data pack): your question and country; your unit of observation and years; what you compare to what; your sources; one check number that you have reproduced; who does what.
Before Lesson 4	Download, log and check your team's data against the publisher's numbers (see your question card). Bring your first descriptive statistics.
Lesson 5	Brief (PDF, two pages), slides , and replication files (zip). One upload per group, before class. In class: your presentation to FZS.

The brief: two pages, four parts

The references and the appendix do not count toward the two pages.

1. **The question** (about half a page). The problem FZS faces, why it matters to them, the question you ask, and the places and years you study.
2. **Data and provenance** (about a quarter page). Name each dataset you use and its source. For each one, report one check: a number from your data that matches a figure the publisher shows.
3. **Findings**. One figure that makes your main point, with a short interpretation. Label every claim as *description*, *prediction* or *causal*. You are not expected to make a causal claim, but you must not make one by accident.
4. **What FZS can do with this**. Three short paragraphs: what FZS can learn from your analysis; why it could mislead (sources of endogeneity and measurement error, and what the satellite data cannot see, such as mercury in rivers); and what data or research design would let you say more.

Appendix: the data log, the references, and the AI output you used for part 4.

How to proceed

1. **Describe**. One table of your outcome by unit and year, in levels and in annual additions (the change from the year before). Required.
2. **Compare**. Two groups of units, two periods: four numbers. Explain your choice of comparison group and of date. Required.
3. **Model**. One regression with unit and year fixed effects, standard errors clustered by unit. Your question card gives the specification. Optional.

A careful description and comparison count for more than a regression that you cannot explain.

Sources and citations

- **Numbers from your own analysis** must be produced by your replication code. If your code does not produce a number, do not report it.
- **Numbers from other sources** are cited in the text (author or organization, year) and listed in the references with title, journal or publisher, link and date of access. For reports, give the page or figure.
- **Data** are cited too: publisher, dataset, collection or version, link, date of download and download settings. This is your data log (DOWNLOAD_LOG.csv).

- **AI.** You may use AI as a coding assistant, and for part 4 with the prompt below. AI output is not a source: do not cite it for facts. AI assistants sometimes invent references, so list only sources that you have opened yourself.
- **Check.** We will pick one number from your brief and trace it back, through your code and your data log, to the publisher.

Prompt for part 4 (include the AI’s answer and your edited version in the appendix): *“Here is my comparison: [units, years, outcome, and what you compare to what]. List the reasons this comparison could mislead, distinguishing omitted variables, reverse causation, selection of units, and measurement error in the outcome. For each, say what data or design would fix it.”*

Replication files

Someone who has not seen your work should be able to unzip your folder, run one file, and obtain your figure and every number in your brief.

In the folder	What it is
README.txt	Software and version; which file to run; what it produces.
data_raw/	Every file you used, exactly as downloaded. Do not edit raw files by hand. Leave out files above 50 MB and give the link in the data log.
DOWNLOAD_LOG.csv	One row per raw file.
code/	One script or notebook that runs from top to bottom, in any software.
output/	Your figure, and the numbers that appear in your brief.

Practical points

- **Countries.** Your analysis uses data for Peru and Brazil. In Colombia most mining takes place on rivers, which land-cover data barely capture, so Colombia is not part of the analysis.
- **No GIS needed.** Every source on your question card is a table (place by year). The publishers have already done the spatial work.
- **What the data cannot see.** River dredging and mercury pollution are major harms of alluvial gold mining. Neither appears in land-cover data. Every brief has to say so.
- **Help.** I am happy to discuss your question, your comparison and what kind of claim you can make. For coding problems, use your AI assistant. You remain responsible for what is estimated and for whether the output makes sense.

Assessment

Your grade is based on your written deliverables: the brief, its appendix and the replication files. The presentation is required, but it is not graded.

Framing and partner perspective (20%); provenance: data log, check against the source, replication files and the “cannot see” paragraph (25%); analysis and honesty about what kind of claim it supports (30%); the figure (10%); would FZS act on it (15%).

Regression model (optional)

For place i in year t , where y is the annual addition to mining area:

$$y_{it} = \alpha_i + \gamma_t + \beta x_{it} + \varepsilon_{it}$$

α_i are place fixed effects (everything about a place that does not change over time). γ_t are year fixed effects (everything that affects all places in the same year, including the gold price). x_{it} has to vary across both places and years, for example *targeted area* $\times$ *after 2019*, or *log gold price* $\times$ *exposure of the place* (for example, the share of its area that was mining in 2010). Cluster the standard errors by place. A few places account for most of the mining area: check whether your result changes when you leave out the largest one.

```
R      fixest::feols(y ~ x | place + year, data = d, cluster = ~place)
Python pyfixest.feols("y ~ x | place + year", data=d, vcov={"CRV1": "place"})
Stata   reghdfe y x, absorb(place year) vce(cluster place)
```

The data pack

`gold_student_pack.zip` contains six tables, a data dictionary (`DATA_DICTIONARY.pdf`: every variable, with its unit), starter code, the data log and templates. Run the starter code first: it loads every table and reproduces the check numbers below.

File (in <code>data/</code>)	One row is	Main variables	Source
<code>prices_annual.csv</code> , <code>prices_monthly.csv</code>	a year (1961 to 2025); a month (to August 2026)	<code>gold</code> (\$ per troy ounce), <code>silver</code> , <code>copper</code> , <code>tin</code> , oil; logs and indices	World Bank Commodity Price Data (“Pink Sheet”), September 2026
<code>peru_department_year.csv</code>	department $\times$ biome $\times$ year, 1985 to 2025	<code>mining_ha</code> , <code>forest_ha</code> , <code>total_ha</code>	MapBiomass Peru, Collection 4
<code>peru_bufferzone_year.csv</code>	buffer zone of a protected area $\times$ department $\times$ year, 1985 to 2025	same, and <code>pa_category</code>	MapBiomass Peru, Collection 4
<code>brazil_municipality_year.csv</code>	municipality $\times$ state $\times$ biome $\times$ year, 1985 to 2024	same, and <code>state_acronym</code>	MapBiomass Brazil, Collection 10.1
<code>mining_area.csv</code>	Brazil as a whole, and all indigenous territories combined, by year, 1985 to 2025	mining area by type (artisanal, industrial) and substance	MapBiomass Brazil platform, Collection 11

Check numbers. Gold price, August 2026: \$4,411 per troy ounce · Brazil, all municipalities, 2024: 609,637 ha · Madre de Dios, 2025: 112,622 ha · Tambopata buffer zone, 2025: 20,730 ha · Brazil, artisanal mining, 2025: 445,987 ha · artisanal mining in all Brazilian indigenous territories, 2025: 39,915 ha.

Notes on the data

Terms. *Artisanal* mining is small-scale mining (in Brazil: *garimpo*). *Alluvial* gold is found in river sediments and floodplains. A *buffer zone* is the area surrounding a protected area. *MAAP* reports are published by the NGO Amazon Conservation (maaprogram.org).

What is measured. MapBiomass classifies satellite images (Landsat, 30-meter pixels) into land-cover classes, for every year. `mining_ha` is the area in the class “mining”. It includes all mining: legal and illegal, artisanal and industrial, gold and other minerals.

- **Peru.** Keep `biome == "Amazonia"`. The other biomes (Andes, coast) contain large industrial mines. Within the Amazon biome, 95% of the mining area in 2025 is in the department of Madre de Dios, where mining is alluvial gold mining.
- **Brazil, municipalities.** The mining class includes industrial mines, for example iron ore in Parauapebas and bauxite in Oriximiná and Paragominas. Explain how you choose your municipalities. Identify a municipality by state and name together. Where a municipality borders another state, the table can contain a second row for it under that state, with a few hectares: drop units with a very small `total_ha`.
- **Brazil, by type and substance.** Only `mining_area.csv` separates artisanal from industrial mining, and gold from other substances. Inside indigenous territories most artisanal mining has no substance recorded (27,875 of 39,915 ha in 2025). The variable `gold_or_nosub_artisanal_ha` adds the two, and is our working measure of illegal gold mining.
- **More than one row per place and year.** A department or municipality has one row per biome, and a buffer zone one row per department. Add them up before you compute anything.

Levels and additions. The tables report the area classified as mining in a given year. This area rarely falls, so it trends upward, as does the gold price. Two series that trend upward are correlated even if they are unrelated. Use annual **additions** (this year’s area minus last year’s). Note that mining on an old mining site that had regrown counts as an addition, and that additions can be negative when old sites are reclassified.

Different sources, different definitions. For Madre de Dios, MapBiomass Peru reports 112,622 ha classified as mining in 2025. MAAP report #233 reports 135,939 ha of cumulative deforestation due to gold mining, as of mid-2025. These numbers measure different things and should not be subtracted from each other.